

Manual Change History (4021442) C8.3 Marine Electronic Control System Troubleshooting and Repair Manual

Date	12-Mar-2002
Section	Section i - Introduction
Procedure Number	99-204-004 (/qs3/pubsys2/xml/en/procedures/99/99-204-004.html)
Procedure Title	Symbols
Update Reason	

Date	12-Mar-2002
Section	Section i - Introduction
Procedure Number	99-204-005 (/qs3/pubsys2/xml/en/procedures/99/99-204-005.html)
Procedure Title	Illustrations
Update Reason	

Date	12-Mar-2002
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-361 (/qs3/pubsys2/xml/en/procedures/99/99-019-361.html)
Procedure Title	Component Connector and Pin Inspection
Update Reason	

Date	03-Apr-2002
Section	Section TS - Troubleshooting Symptoms
Procedure Number	99-t00-002 (/qs3/pubsys2/xml/en/procedures/99/99-t00-002.html)
Procedure Title	Troubleshooting Symptoms Charts
Update Reason	

Date	10-Jul-2002
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-220 (/qs3/pubsys2/xml/en/procedures/99/99-019-220.html)
Procedure Title	Augat Connector Series
Update Reason	
Date	22-Jan-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-052 (/qs3/pubsys2/xml/en/procedures/99/99-019-052.html)
Procedure Title	Idle Adjust Switch
Update Reason	
Date	03-Apr-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-027 (/qs3/pubsys2/xml/en/procedures/99/99-019-027.html)
Procedure Title	Diagnostic Test Mode Switch
Update Reason	
Date	03-Apr-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-046 (/qs3/pubsys2/xml/en/procedures/99/99-019-046.html)
Procedure Title	Fault Lamp
Update Reason	

Date	03-Apr-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-085 (/qs3/pubsys2/xml/en/procedures/99/99-019-085.html)
Procedure Title	Accelerator Pedal or Lever Position Sensor
Update Reason	
Date	03-Apr-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-197 (/qs3/pubsys2/xml/en/procedures/99/99-019-197.html)
Procedure Title	Ring Terminal
Update Reason	
Date	03-Apr-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-198 (/qs3/pubsys2/xml/en/procedures/99/99-019-198.html)
Procedure Title	Fuse, Harness In-Line
Update Reason	
Date	03-Apr-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-199 (/qs3/pubsys2/xml/en/procedures/99/99-019-199.html)
Procedure Title	Connector, Butt Splice
Update Reason	

Date	03-Apr-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-202 (/qs3/pubsys2/xml/en/procedures/99/99-019-202.html)
Procedure Title	Metripack Connector Series
Update Reason	
Date	03-Apr-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-203 (/qs3/pubsys2/xml/en/procedures/99/99-019-203.html)
Procedure Title	AMP Connector Series
Update Reason	
Date	03-Apr-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-204 (/qs3/pubsys2/xml/en/procedures/99/99-019-204.html)
Procedure Title	Deutsch™ DRC Connector Series
Update Reason	
Date	03-Apr-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-206 (/qs3/pubsys2/xml/en/procedures/99/99-019-206.html)
Procedure Title	Deutsch™ DTM and DTP Connector Series
Update Reason	

Date	03-Apr-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-209 (/qs3/pubsys2/xml/en/procedures/99/99-019-209.html)
Procedure Title	ITT Cannon Connector Series
Update Reason	
Date	09-Oct-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-201 (/qs3/pubsys2/xml/en/procedures/99/99-019-201.html)
Procedure Title	Weather Pak Connector Series
Update Reason	
Date	09-Oct-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-205 (/qs3/pubsys2/xml/en/procedures/99/99-019-205.html)
Procedure Title	Deutsch™ DT Connector Series
Update Reason	
Date	09-Oct-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-207 (/qs3/pubsys2/xml/en/procedures/99/99-019-207.html)
Procedure Title	Deutsch™ HD10 Connector Series
Update Reason	

Date	09-Oct-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-210 (/qs3/pubsys2/xml/en/procedures/99/99-019-210.html)
Procedure Title	AMP VP44 Connector Series
Update Reason	
Date	09-Oct-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-211 (/qs3/pubsys2/xml/en/procedures/99/99-019-211.html)
Procedure Title	Packard Relay Connector
Update Reason	
Date	09-Oct-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-212 (/qs3/pubsys2/xml/en/procedures/99/99-019-212.html)
Procedure Title	Ford™ Connector Series
Update Reason	
Date	09-Oct-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-213 (/qs3/pubsys2/xml/en/procedures/99/99-019-213.html)
Procedure Title	D-Sub Miniature Connector Series
Update Reason	

Date	09-Oct-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-218 (/qs3/pubsys2/xml/en/procedures/99/99-019-218.html)
Procedure Title	Framatome Connector Series
Update Reason	
Date	29-Oct-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	07-fc121 (/qs3/pubsys2/xml/en/procedures/07/07-fc121.html)
Procedure Title	Engine Speed Sensor - Lost Secondary Engine Speed Sensor
Update Reason	
Date	29-Oct-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	07-fc124 (/qs3/pubsys2/xml/en/procedures/07/07-fc124.html)
Procedure Title	Intake Manifold Pressure High - Warning
Update Reason	
Date	29-Oct-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	07-fc132 (/qs3/pubsys2/xml/en/procedures/07/07-fc132.html)
Procedure Title	Low Voltage Detected at the Throttle Position Signal Circuit
Update Reason	

Date	29-Oct-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	07-fc133 (/qs3/pubsys2/xml/en/procedures/07/07-fc133.html)
Procedure Title	Backup Throttle Position Sensor - Shorted High
Update Reason	
Date	29-Oct-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	07-fc134 (/qs3/pubsys2/xml/en/procedures/07/07-fc134.html)
Procedure Title	Backup Throttle Position Sensor - Shorted Low
Update Reason	
Date	29-Oct-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	07-fc135 (/qs3/pubsys2/xml/en/procedures/07/07-fc135.html)
Procedure Title	High Pressure Detected at the Oil Pressure Circuit
Update Reason	
Date	03-Nov-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	07-fc111 (/qs3/pubsys2/xml/en/procedures/07/07-fc111.html)
Procedure Title	Electronic Control Module Critical Internal Failure - Bad Intelligent Device or Component
Update Reason	

Date	03-Nov-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	07-fc141 (/qs3/pubsys2/xml/en/procedures/07/07-fc141.html)
Procedure Title	Low Voltage Detected at the Oil Pressure Circuit
Update Reason	
Date	03-Nov-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	07-fc143 (/qs3/pubsys2/xml/en/procedures/07/07-fc143.html)
Procedure Title	Oil Pressure Signal Indicates Oil Pressure Below the Low Oil Pressure Engine Protection Limit
Update Reason	
Date	03-Nov-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	07-fc144 (/qs3/pubsys2/xml/en/procedures/07/07-fc144.html)
Procedure Title	High Voltage Detected at the Coolant Temperature Circuit
Update Reason	
Date	03-Nov-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	07-fc145 (/qs3/pubsys2/xml/en/procedures/07/07-fc145.html)
Procedure Title	Low Voltage Detected at the Coolant Temperature Circuit
Update Reason	

Date	03-Nov-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	07-fc151 (/qs3/pubsys2/xml/en/procedures/07/07-fc151.html)
Procedure Title	Coolant Temperature Signal Indicates Coolant Temperature Above 104°C [220°F]
Update Reason	
Date	03-Nov-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	07-fc153 (/qs3/pubsys2/xml/en/procedures/07/07-fc153.html)
Procedure Title	High Voltage Detected at the Intake Manifold Temperature Circuit
Update Reason	
Date	03-Nov-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	07-fc154 (/qs3/pubsys2/xml/en/procedures/07/07-fc154.html)
Procedure Title	Low Voltage Detected at the Intake Manifold Temperature Circuit
Update Reason	
Date	03-Nov-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	07-fc155 (/qs3/pubsys2/xml/en/procedures/07/07-fc155.html)
Procedure Title	Intake Manifold Temperature Signal Indicates Temperature Above 88°C [190°F]
Update Reason	

Date	03-Nov-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	07-fc166 (/qs3/pubsys2/xml/en/procedures/07/07-fc166.html)
Procedure Title	The Rack Position Sensor Indicates Rack Position is Greater than the Calibrated Threshold
Update Reason	

Date	01-Dec-2003
Section	Section E - Engine and System Identification
Procedure Number	07-100-001 (/qs3/pubsys2/xml/en/procedures/07/07-100-001.html)
Procedure Title	Engine Identification
Update Reason	

Date	01-Dec-2003
Section	Section F - Familiarization
Procedure Number	07-101-007 (/qs3/pubsys2/xml/en/procedures/07/07-101-007.html)
Procedure Title	Electronic Controlled Fuel System
Update Reason	

Date	01-Dec-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	07-fcmfc01 (/qs3/pubsys2/xml/en/procedures/07/07-fcmfc01.html)
Procedure Title	Fault Code Multiple A
Update Reason	

Date	01-Dec-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	07-fcmfc02 (/qs3/pubsys2/xml/en/procedures/07/07-fcmfc02.html)
Procedure Title	Fault Code Multiple B
Update Reason	

Date	01-Dec-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	07-fc172 (/qs3/pubsys2/xml/en/procedures/07/07-fc172.html)
Procedure Title	Fuel Rack is Stuck in a Position Commanding Excessive Fueling to the Engine
Update Reason	
Date	01-Dec-2003
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	07-fc173 (/qs3/pubsys2/xml/en/procedures/07/07-fc173.html)
Procedure Title	Fuel Rack is Stuck in a Position Providing Adequate or Less than Adequate Fueling to the Engine
Update Reason	
Date	01-Dec-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	07-022-001_19 (/qs3/pubsys2/xml/en/procedures/07/07-022-001_19.html)
Procedure Title	Service Tools
Update Reason	
Date	01-Dec-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	07-019-059 (/qs3/pubsys2/xml/en/procedures/07/07-019-059.html)
Procedure Title	Intake Manifold Air Temperature Sensor
Update Reason	

Date	01-Dec-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	07-019-061 (/qs3/pubsys2/xml/en/procedures/07/07-019-061.html)
Procedure Title	Intake Manifold Pressure Sensor
Update Reason	
Date	01-Dec-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	07-019-066 (/qs3/pubsys2/xml/en/procedures/07/07-019-066.html)
Procedure Title	Engine Oil Pressure Sensor
Update Reason	
Date	01-Dec-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	07-019-087 (/qs3/pubsys2/xml/en/procedures/07/07-019-087.html)
Procedure Title	Unswitched Battery Supply Circuit
Update Reason	
Date	01-Dec-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	07-019-106 (/qs3/pubsys2/xml/en/procedures/07/07-019-106.html)
Procedure Title	Engine Speed Sensor Circuit
Update Reason	

Date	01-Dec-2003
Section	Section L - Service Literature
Procedure Number	07-205-001 (/qs3/pubsys2/xml/en/procedures/07/07-205-001.html)
Procedure Title	Additional Service Literature
Update Reason	
Date	02-Dec-2003
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021442-t081-tr (/qs3/pubsys2/xml/en/manual/4021442/4021442-t081-tr.html)
Procedure Title	Engine Will Not Shut Off
Update Reason	
Date	02-Dec-2003
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021442-t083-tr (/qs3/pubsys2/xml/en/manual/4021442/4021442-t083-tr.html)
Procedure Title	Fault Code Warning Lamps Stay On (No Apparent Reason)
Update Reason	
Date	02-Dec-2003
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021442-t084-tr (/qs3/pubsys2/xml/en/manual/4021442/4021442-t084-tr.html)
Procedure Title	Fault Code Warning Lamps Do Not Illuminate
Update Reason	

Date	02-Dec-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	07-019-086 (/qs3/pubsys2/xml/en/procedures/07/07-019-086.html)
Procedure Title	Accelerator Pedal or Lever Position Sensor Circuit
Update Reason	

Date	09-Dec-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	07-019-031 (/qs3/pubsys2/xml/en/procedures/07/07-019-031.html)
Procedure Title	Electronic Control Module (ECM)
Update Reason	

Date	09-Dec-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	07-019-032 (/qs3/pubsys2/xml/en/procedures/07/07-019-032.html)
Procedure Title	ECM Calibration Code
Update Reason	

Date	09-Dec-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	07-019-042 (/qs3/pubsys2/xml/en/procedures/07/07-019-042.html)
Procedure Title	Engine Speed Sensor (ESS)
Update Reason	

Date	09-Dec-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	07-019-043 (/qs3/pubsys2/xml/en/procedures/07/07-019-043.html)
Procedure Title	Engine Wiring Harness
Update Reason	
Date	09-Dec-2003
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	07-019-050 (/qs3/pubsys2/xml/en/procedures/07/07-019-050.html)
Procedure Title	Fuel Shutoff Valve
Update Reason	
Date	23-Jan-2004
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021442-t061-tr (/qs3/pubsys2/xml/en/manual/4021442/4021442-t061-tr.html)
Procedure Title	Engine Runs Rough at Idle
Update Reason	
Date	23-Jan-2004
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021442-t062-tr (/qs3/pubsys2/xml/en/manual/4021442/4021442-t062-tr.html)
Procedure Title	Engine Runs Rough or Misfires
Update Reason	

Date	23-Jan-2004
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021442-t064-tr (/qs3/pubsys2/xml/en/manual/4021442/4021442-t064-tr.html)
Procedure Title	Engine Shuts Off Unexpectedly or Dies During Deceleration
Update Reason	
Date	23-Jan-2004
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021442-t066-tr (/qs3/pubsys2/xml/en/manual/4021442/4021442-t066-tr.html)
Procedure Title	Engine Speed Surges at Low or High Idle
Update Reason	
Date	23-Jan-2004
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021442-t080-tr (/qs3/pubsys2/xml/en/manual/4021442/4021442-t080-tr.html)
Procedure Title	Engine Will Not Reach Rated Speed (RPM)
Update Reason	
Date	23-Jan-2004
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021442-t118-tr (/qs3/pubsys2/xml/en/manual/4021442/4021442-t118-tr.html)
Procedure Title	Smoke, White – Excessive
Update Reason	
Date	03-Mar-2004
Section	Section V - Specifications
Procedure Number	99-018-010 (/qs3/pubsys2/xml/en/procedures/99/99-018-010.html)
Procedure Title	Fraction, Decimal, Millimeter Conversions
Update Reason	

Date	03-Mar-2004
Section	Section V - Specifications
Procedure Number	99-018-011 (/qs3/pubsys2/xml/en/procedures/99/99-018-011.html)
Procedure Title	Newton-Meter to Foot-Pound Conversions
Update Reason	
Date	03-Mar-2004
Section	Section V - Specifications
Procedure Number	99-018-012 (/qs3/pubsys2/xml/en/procedures/99/99-018-012.html)
Procedure Title	Pipe Plug Torque Values
Update Reason	
Date	03-Mar-2004
Section	Section V - Specifications
Procedure Number	99-018-013 (/qs3/pubsys2/xml/en/procedures/99/99-018-013.html)
Procedure Title	Tap-Drill Chart - U.S. Customary and Metric
Update Reason	
Date	03-Mar-2004
Section	Section V - Specifications
Procedure Number	99-018-014 (/qs3/pubsys2/xml/en/procedures/99/99-018-014.html)
Procedure Title	Weights and Measures - Conversion Factors
Update Reason	
Date	15-Mar-2004
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	07-019-019 (/qs3/pubsys2/xml/en/procedures/07/07-019-019.html)
Procedure Title	Engine Coolant Temperature Sensor
Update Reason	

Date	19-Mar-2004
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	07-fc122 (/qs3/pubsys2/xml/en/procedures/07/07-fc122.html)
Procedure Title	Intake Manifold Pressure Sensor Circuit - Voltage Above Normal or Shorted to High Source
Update Reason	

Date	19-Mar-2004
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	07-fc123 (/qs3/pubsys2/xml/en/procedures/07/07-fc123.html)
Procedure Title	Intake Manifold Pressure Sensor Circuit - Voltage Below Normal or Shorted to Low Source
Update Reason	

Date	19-Mar-2004
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	07-fc131 (/qs3/pubsys2/xml/en/procedures/07/07-fc131.html)
Procedure Title	High Voltage Detected at the Throttle Position Signal Circuit
Update Reason	

Date	19-Mar-2004
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	07-019-049 (/qs3/pubsys2/xml/en/procedures/07/07-019-049.html)
Procedure Title	Fuel Shutoff Valve Circuit
Update Reason	

Date	30-Mar-2004
Section	front
Procedure Number	4021442-toc (/qs3/pubsys2/xml/en/manual/4021442/4021442-toc.html)
Procedure Title	table of contents
Update Reason	
Date	02-Jun-2004
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021442-t033-tr (/qs3/pubsys2/xml/en/manual/4021442/4021442-t033-tr.html)
Procedure Title	Engine Acceleration or Response Poor
Update Reason	
Date	02-Jun-2004
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021442-t043-tr (/qs3/pubsys2/xml/en/manual/4021442/4021442-t043-tr.html)
Procedure Title	Engine Difficult to Start or Will Not Start (Exhaust Smoke)
Update Reason	
Date	02-Jun-2004
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021442-t044-tr (/qs3/pubsys2/xml/en/manual/4021442/4021442-t044-tr.html)
Procedure Title	Engine Difficult to Start or Will Not Start (No Exhaust Smoke)
Update Reason	
Date	02-Jun-2004
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021442-t057-tr (/qs3/pubsys2/xml/en/manual/4021442/4021442-t057-tr.html)
Procedure Title	Engine Power Output Low
Update Reason	

Date	02-Jun-2004
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021442-t067-tr (/qs3/pubsys2/xml/en/manual/4021442/4021442-t067-tr.html)
Procedure Title	Engine Speed Surges Under Load or in Operating Range
Update Reason	
Date	02-Jun-2004
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021442-t072-tr (/qs3/pubsys2/xml/en/manual/4021442/4021442-t072-tr.html)
Procedure Title	Engine Starts But Will Not Keep Running
Update Reason	
Date	02-Jun-2004
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021442-t087-tr (/qs3/pubsys2/xml/en/manual/4021442/4021442-t087-tr.html)
Procedure Title	Fuel Consumption Excessive
Update Reason	
Date	02-Jun-2004
Section	Section TS - Troubleshooting Symptoms
Procedure Number	4021442-t116-tr (/qs3/pubsys2/xml/en/manual/4021442/4021442-t116-tr.html)
Procedure Title	Smoke, Black – Excessive
Update Reason	
Date	09-Jun-2004
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-208 (/qs3/pubsys2/xml/en/procedures/99/99-019-208.html)
Procedure Title	Deutsch™ HDP20 and HD30 Connector Series
Update Reason	

Date	02-Nov-2004
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	07-fc115 (/qs3/pubsys2/xml/en/procedures/07/07-fc115.html)
Procedure Title	Engine Speed Sensor - Lost Primary Engine Speed Sensor
Update Reason	
Date	10-Mar-2005
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	07-fc497 (/qs3/pubsys2/xml/en/procedures/07/07-fc497.html)
Procedure Title	Multi-Unit Sync Switch Error
Update Reason	
Date	06-Feb-2006
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	105-019-428 (/qs3/pubsys2/xml/en/procedures/105/105-019-428.html)
Procedure Title	Engine Datalinks
Update Reason	
Date	28-Feb-2006
Section	Section V - Specifications
Procedure Number	4021331 (/qs3/pubsys2/xml/en/procedures//4021331.html)
Procedure Title	Specifications
Update Reason	
Date	28-Feb-2006
Section	back
Procedure Number	4021442-backcover (/qs3/pubsys2/xml/en/manual/4021442/4021442-backcover.html)
Procedure Title	Backcover
Update Reason	

Date	02-Mar-2006
Section	front
Procedure Number	4021442-titlepage (/qs3/pubsys2/xml/en/manual/4021442/4021442-titlepage.html)
Procedure Title	titlepage
Update Reason	
Date	02-Mar-2006
Section	Section E - Engine and System Identification
Procedure Number	07-100-002-tr (/qs3/pubsys2/xml/en/procedures/07/07-100-002-tr.html)
Procedure Title	Engine Diagrams
Update Reason	
Date	22-May-2006
Section	front
Procedure Number	tr-frwd (/qs3/pubsys2/xml/en/manual/preface/tr-frwd.html)
Procedure Title	foreword
Update Reason	
Date	27-Jun-2006
Section	Section L - Service Literature
Procedure Number	99-205-002 (/qs3/pubsys2/xml/en/procedures/99/99-205-002.html)
Procedure Title	Service Literature Ordering Location
Update Reason	
Date	23-Jul-2007
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	07-fc237 (/qs3/pubsys2/xml/en/procedures/07/07-fc237.html)
Procedure Title	Multi-Unit Sync Error
Update Reason	

Date	24-Jul-2007
Section	Section i - Introduction
Procedure Number	99-204-008 (/qs3/pubsys2/xml/en/procedures/99/99-204-008.html)
Procedure Title	General Cleaning Instructions
Update Reason	
Date	07-Dec-2007
Section	Section TT- Troubleshooting Symptoms (New Format)
Procedure Number	00-ttecm (/qs3/pubsys2/xml/en/procedures/00/00-ttecm.html)
Procedure Title	ECM – No Communication Troubleshooting Tree
Update Reason	
Date	28-Jan-2008
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-360 (/qs3/pubsys2/xml/en/procedures/99/99-019-360.html)
Procedure Title	Resistance Measurement Using a Multimeter and a Wiring Diagram
Update Reason	
Date	30-May-2008
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-008 (/qs3/pubsys2/xml/en/procedures/99/99-019-008.html)
Procedure Title	Battery Ground Circuit
Update Reason	

Date	30-May-2008
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-028 (/qs3/pubsys2/xml/en/procedures/99/99-019-028.html)
Procedure Title	Diagnostic Test Mode Switch Circuit
Update Reason	
Date	30-May-2008
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-047 (/qs3/pubsys2/xml/en/procedures/99/99-019-047.html)
Procedure Title	Fault Lamp Circuit
Update Reason	
Date	30-May-2008
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-053 (/qs3/pubsys2/xml/en/procedures/99/99-019-053.html)
Procedure Title	Idle Adjust Switch Circuit
Update Reason	
Date	30-May-2008
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-166 (/qs3/pubsys2/xml/en/procedures/99/99-019-166.html)
Procedure Title	Data Link Circuit, SAE J1587
Update Reason	

Date	11-Jun-2008
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-165 (/qs3/pubsys2/xml/en/procedures/99/99-019-165.html)
Procedure Title	Data Link Circuit, SAE J1939
Update Reason	
Date	19-Mar-2009
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-215 (/qs3/pubsys2/xml/en/procedures/99/99-019-215.html)
Procedure Title	Bosch® ECM Injector Driver Connector Series
Update Reason	
Date	23-Apr-2009
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-216 (/qs3/pubsys2/xml/en/procedures/99/99-019-216.html)
Procedure Title	Bosch® ECM Actuator and Sensor Connector Series
Update Reason	
Date	23-Apr-2009
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-217 (/qs3/pubsys2/xml/en/procedures/99/99-019-217.html)
Procedure Title	Bosch® ECM OEM Connector Series
Update Reason	

Date	25-Jun-2009
Section	Section TS - Troubleshooting Symptoms
Procedure Number	99-t00-001-tr (/qs3/pubsys2/xml/en/procedures/99/99-t00-001-tr.html)
Procedure Title	Troubleshooting Procedures and Techniques
Update Reason	
Date	22-Jul-2009
Section	Section L - Service Literature
Procedure Number	99-205-004 (/qs3/pubsys2/xml/en/procedures/99/99-205-004.html)
Procedure Title	Cummins Customized Parts Catalog
Update Reason	
Date	10-Sep-2009
Section	Section i - Introduction
Procedure Number	99-204-002-tr (/qs3/pubsys2/xml/en/procedures/99/99-204-002-tr.html)
Procedure Title	About the Manual
Update Reason	
Date	16-Sep-2009
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-064 (/qs3/pubsys2/xml/en/procedures/99/99-019-064.html)
Procedure Title	Key Switch Battery Supply Circuit
Update Reason	
Date	08-Oct-2009
Section	Section i - Introduction
Procedure Number	99-204-007-tr (/qs3/pubsys2/xml/en/procedures/99/99-204-007-tr.html)
Procedure Title	General Repair Instructions
Update Reason	

Date	02-Dec-2009
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-359 (/qs3/pubsys2/xml/en/procedures/99/99-019-359.html)
Procedure Title	Multimeter Usage
Update Reason	
Date	02-Dec-2009
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	99-019-362 (/qs3/pubsys2/xml/en/procedures/99/99-019-362.html)
Procedure Title	Inactive or Intermittent Fault Code
Update Reason	
Date	25-Jan-2010
Section	Section i - Introduction
Procedure Number	99-204-006-tr (/qs3/pubsys2/xml/en/procedures/99/99-204-006-tr.html)
Procedure Title	General Safety Instructions
Update Reason	
Date	25-Jan-2010
Section	Section V - Specifications
Procedure Number	99-018-005 (/qs3/pubsys2/xml/en/procedures/99/99-018-005.html)
Procedure Title	Drive Belt Tension
Update Reason	
Date	25-Jan-2010
Section	Section V - Specifications
Procedure Number	99-018-009 (/qs3/pubsys2/xml/en/procedures/99/99-018-009.html)
Procedure Title	Capscrew Markings and Torque Values
Update Reason	

Date	18-Feb-2010
Section	Section F - Familiarization
Procedure Number	00-022-999 (/qs3/pubsys2/xml/en/procedures/00/00-022-999.html)
Procedure Title	Service Tools and Hardware - Overview
Update Reason	
Date	24-Mar-2010
Section	Section i - Introduction
Procedure Number	99-204-009 (/qs3/pubsys2/xml/en/procedures/99/99-204-009.html)
Procedure Title	Acronyms and Abbreviations
Update Reason	
Date	25-May-2010
Section	Section i - Introduction
Procedure Number	99-204-003-tr (/qs3/pubsys2/xml/en/procedures/99/99-204-003-tr.html)
Procedure Title	How to Use the Manual
Update Reason	
Date	21-Jan-2011
Section	Section F - Familiarization
Procedure Number	00-022-999 (/qs3/pubsys2/xml/en/procedures/00/00-022-999.html)
Procedure Title	Service Tools and Hardware - Overview
Update Reason	Updated General Information, Initial Check, Setup, and Resistance Check sections.
Date	03-Jul-2013
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	4021442-t087-tr (/qs3/pubsys2/xml/en/manual/4021442/4021442-t087-tr.html)
Procedure Title	Fuel Consumption Excessive
Update Reason	Removed Cause: Fuel economy expectations are not reasonable

Date	07-Oct-2016
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	07-fc346 (/qs3/pubsys2/xml/en/procedures/07/07-fc346.html)
Procedure Title	Engine Control Module Calibration Memory Software - Bad Intelligent Device or Component
Update Reason	Added to manual.
Date	07-Oct-2016
Section	Section TF - Troubleshooting Fault Codes
Procedure Number	07-t05-346 (/qs3/pubsys2/xml/en/procedures/07/07-t05-346.html)
Procedure Title	Engine Control Module Calibration Memory Software - Bad Intelligent Device or Component
Update Reason	Added to manual.
Date	05-Dec-2016
Section	Section TT - Troubleshooting Symptoms (New Format)
Procedure Number	00-ttecm (/qs3/pubsys2/xml/en/procedures/00/00-ttecm.html)
Procedure Title	ECM - No Communication Troubleshooting Tree
Update Reason	Modified Steps 2, 6, 9 and 11 and added information for new INLINE™ 7 data link.
Date	13-Dec-2016
Section	Section F - Familiarization
Procedure Number	00-022-999 (/qs3/pubsys2/xml/en/procedures/00/00-022-999.html)
Procedure Title	Service Tools and Hardware - Overview
Update Reason	Removed or corrected information for INLINE™ 4, 5, and 6. Added new information for INLINE™ 7 data link.

Date	24-Sep-2024
Section	Section 19 - Electronic Engine Controls - Group 19
Procedure Number	07-019-031 (/qs3/pubsys2/xml/en/procedures/07/07-019-031.html)
Procedure Title	Engine Control Module
Update Reason	No changes made to procedure, needed to update procedure title to align to proper assigned title.

Last Modified: 03-Feb-2025
